

THE MUMFORD–TATE GROUP OF A VERY GENERAL LAGRANGIAN FIBER

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ABSTRACT. For a primitive symplectic variety X admitting a holomorphic Lagrangian fibration $f : X \rightarrow B$, we fully classify the possible Mumford–Tate groups of the very general fiber. In the special case when X is a general hyper-Kähler manifold with $b_2(X) \geq 7$, we further show that the very general fiber is Hodge-generic. We extract as a consequence that the discriminant of f has codimension one in B . Finally, as an application of our classification, we confirm an expectation of Li–Tosatti that when f is not isotrivial, the special Kähler metric on B has positive holomorphic sectional curvature on the regular values of f .

1. INTRODUCTION

In this paper, we will use Hodge-theoretic methods to study holomorphic symplectic varieties. We will start with the following definitions.

Definition 1.1. Let X and B be normal analytic varieties, and $f : X \rightarrow B$ a non-trivial surjective morphism with connected fibers.

- (1) We say that X is **holomorphic symplectic** if it admits a non-degenerate 2-form $\sigma \in H^0(X^{\text{sm}}, \Omega_{X^{\text{sm}}}^2)$ which extends holomorphically onto a resolution of singularities $Y \rightarrow X$.
- (2) We say that a subvariety $Z \subseteq X$ is **Lagrangian** if $\sigma|_Z = 0$ and $\dim Z = \frac{1}{2} \dim X$. We say that $f : X \rightarrow B$ is a Lagrangian fibration if every fiber of f is Lagrangian.
- (3) Suppose that X is compact, $\mathbb{Q}$ -factorial, and log-terminal. We say that X is **primitive symplectic** if $H^1(X, \mathcal{O}_X) = 0$ and the symplectic form σ is unique up to scaling.
- (4) If X is a compact, simply connected Kähler manifold with a symplectic form σ unique up to scaling, we say that X is **hyper-Kähler**.

By the Beauville–Bogomolov decomposition theorem, any smooth compact symplectic variety is isogenous to a product of hyper-Kähler manifolds and complex tori. Their work has more recently been generalized to compact symplectic varieties, which are isogenous to products of complex tori and special classes of primitive symplectic varieties (see, for instance, [HP19, Theorem 1.5]).

When X is hyper-Kähler, much is understood. For instance, any non-trivial surjective morphism $f : X \rightarrow B$ with connected fibers is forced to be a Lagrangian fibration [GL14], and B is smooth if and only if $B \simeq \mathbb{P}^n$ [Hwa08]. A conjecture of Matsushita predicts that this is always the case.

More generally, suppose X is any primitive symplectic variety admitting a Lagrangian fibration $f : X \rightarrow B$. We abuse notation to denote $f : X^\circ \rightarrow B^\circ$ the restriction of f to the preimage of the regular values, over which the fibers of f are smooth. It turns out that every smooth fiber X_b is an Abelian variety, due to Matsushita [Mat99] when X is projective hyper-Kähler and [Cam06] in more general settings.

Take X to have dimension $2g$, so that both the base B and the fibers X_b are g -dimensional. We denote by $\mathcal{P} : B^\circ \rightarrow \mathcal{A}_{g,\eta}$ the period map associated to f , where $\mathcal{A}_{g,\eta}$ is a moduli space for appropriately-polarized Abelian varieties. If $\mathcal{P}$ is non-constant, that is, if f is not an isotrivial fibration, then $\mathcal{P}$ is generically injective; this is due to [GV16] for X a very general hyper-Kähler and [Bak25] for primitive symplectic X . In the non-isotrivial case, we say that f varies maximally.

Our main technical result provides some description of the image of $\mathcal{P}$. Since $\mathcal{A}_{g,\eta}$ is a Shimura variety; explicitly,

$$\mathcal{A}_{g,\eta} = \Gamma \backslash \mathrm{Sp}_{2g}(\mathbb{R}) / U(g) ,$$

some “special” subvarieties of $\mathcal{A}_{g,\eta}$ are induced by algebraic subgroups $G < \mathrm{Sp}_{2g}$. This subgroup is the Mumford–Tate group of the very general fiber of f . Our result is a full classification of Mumford–Tate groups which can occur in a Lagrangian fibration’s very general fiber.

Theorem 1.2 (=Theorem 4.11). *Let X be a primitive symplectic variety of dimension $2g$, and let $f : X \rightarrow B$ be a Lagrangian fibration. The Mumford–Tate group of f has the form*

$$\mathrm{MT}(X_b) = \begin{cases} (\mathbb{G}_m \times \mathrm{Sp}_{2g/k}^k) / \{\pm 1\} & : f \text{ varies maximally,} \\ \mathbb{G}_m & : f \text{ is isotrivial with fiber isogenous to } E^g \\ & \text{for } E \text{ a CM elliptic curve,} \\ \mathrm{GL}_2 & : \text{otherwise.} \end{cases}$$

There is a more geometric description of this result, which we provide as an obvious corollary. In particular, this generalizes [Mat99] and [Cam06] to describe the structure of the fibers.

Corollary 1.3. *Suppose that $f : X \rightarrow B$ is non-isotrivial. Then for some positive integer k dividing g , the very general fiber X_b is isogenous to a product $A_1 \times \cdots \times A_k$, where the A_i are Hodge-generic Abelian varieties of dimension g/k , which are not pairwise isogenous. In particular, each A_i is simple and has no extra endomorphisms.*

As a direct consequence of this classification, we answer a question of Li–Tosatti. Their result [LT24, Corollary 2.2] shows that if f is not isotrivial, the special Kähler metric ω_{SK} on B° has positive Ricci curvature. In a later remark, they state their stronger expectation that ω_{SK} should have positive holomorphic sectional curvature on B° . The following result proves them right.

Proposition 1.4 (=Theorem 4.12). *Let X be primitive symplectic, and $f : X \rightarrow B$ be a Lagrangian fibration with period map $\mathcal{P} : B^\circ \rightarrow \mathcal{A}_{g,\eta}$ and period matrix $\mathbf{T}$, written as*

$$\mathcal{P}(b) = \begin{pmatrix} \mathbf{T}(b) \\ I_g \end{pmatrix}, \quad \mathbf{T} = \begin{pmatrix} \tau_{11} & \cdots & \tau_{1g} \\ \vdots & \ddots & \vdots \\ \tau_{1g} & \cdots & \tau_{gg} \end{pmatrix}$$

Suppose that $\mathbf{T}$ has a constant diagonal entry τ_{ii} in some analytic neighborhood $U \subseteq B^\circ$. Then f is isotrivial.

By [LT24, Remark 2.3], ω_{SK} having positive holomorphic sectional curvature is equivalent to $\mathbf{T}$ having non-constant diagonal entries, as needed.

Corollary 1.5. *If f is not isotrivial, then ω_{SK} has positive holomorphic sectional curvature on B° .*

It turns out that the deformation space of a polarized hyper-Kähler manifold (X, H) is a Shimura variety. We can construct it as follows: take $L = H^\perp \subsetneq H^2(X, \mathbb{Z})$ with respect to the Beauville–Bogomolov–Fujiki form q , and define

$$\mathcal{D}_H = \{[\sigma] \in \mathbb{P}L_{\mathbb{C}} \mid q(\sigma, \sigma) = 0, \quad q(\sigma, \bar{\sigma}) > 0\}.$$

Then the deformation space of X is constructed as an open subset

$$\mathcal{X} \subseteq \Gamma \backslash \mathcal{D}_H = \Gamma \backslash \left(SO(2, b_2 - 3) / SO(2) \times SO(b_2 - 3) \right),$$

where $b_2 = \dim H^2(X, \mathbb{C})$.

As such, to construct a deformation space of hyper-Kähler manifold, it suffices to exhibit one member. There are currently four known deformation types: the infinite families $K3^{[g]}$ and Kum_g , occurring in every even dimension $2g$, and the two sporadic types $OG6$ and $OG10$, occurring in dimensions 6 and 10 respectively.

Every known deformation space $\mathcal{X}$ contains a member X admitting a Lagrangian fibration $f : X \rightarrow B$. Furthermore, the hyper-Kähler SYZ conjecture predicts that such a member exists in every deformation class of hyper-Kähler manifolds. Matsushita subsequently shows that a codimension one collection of deformations X' of X preserve the Lagrangian fibration f [Mat16].

Thus, we have a notion of X being general among the members of its deformation class admitting a Lagrangian fibration. Our next result strengthens Theorem 4.11 in this setting, concluding that the very general fiber X_b is Hodge-generic.

Theorem 1.6 (=Theorem 5.3). *Let $\mathcal{X}$ be a deformation class of hyper-Kähler manifolds satisfying*

- (1) *there exists $X \in \mathcal{X}$ a hyper-Kähler manifold admitting a Lagrangian fibration,*
- (2) *the second Betti number of any member $X \in \mathcal{X}$ is at least 7.*

Choose $X \in \mathcal{X}$ general among those admitting Lagrangian fibrations, $f : X \rightarrow B$. Then the Mumford–Tate group of the very general fiber X_b is GSp_{2g} .

Once again, it is an obvious consequence that the very general fiber X_b is simple with no extra endomorphisms.

We prove these results by analyzing the monodromy of the variation of Hodge structures underlying f . In the process of proving Theorem 4.11, we also produce the following consequence.

Theorem 1.7 (=Theorem 4.2). *Let $f : X \rightarrow B$ be a Lagrangian fibration, and let $D \subseteq B$ be its discriminant. Then D has codimension one.*

To the best of our knowledge, this was not previously known for primitive symplectic varieties. Hwang [Hwa08, Proposition 4.1] proved this result when X is a projective hyper-Kähler, but his argument does not actually need X to be projective. However, it relies on B being simply connected of Picard rank 1, and therefore fails for primitive symplectic varieties.

In §2, we review preliminaries on variations of weight one Hodge structures, namely Deligne’s canonical extension and Mumford–Tate groups. In §3, we prove a crucial technical

remark of [BS26] about how Lagrangian fibrations degenerate in codimension one. Then §4 proves the classification theorem Theorem 4.11, as well as the discriminant result, Theorem 4.2, and the application to Li–Tosatti, Theorem 4.12. Finally, §5 studies the general deformation of a Lagrangian fibration and proves Theorem 5.3.

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2. PRELIMINARIES

2.1. Deligne’s canonical extension. Let $f : X \rightarrow B$ be a family of Abelian varieties, $D \subseteq B$ the discriminant locus of f , $B^\circ = B \setminus D$, and $f : X^\circ \rightarrow B^\circ$ the smooth part of f , as above. We may consider the underlying $\mathbb{Z}$ -variation of Hodge structures $V = R^1 f_* \mathbb{Z}_{X^\circ}$. Let $\mathcal{V} = V \otimes_{\mathbb{Z}} \mathcal{O}_{B^\circ}$ be the underlying flat bundle, equipped with the flat connection $\nabla : \mathcal{V} \rightarrow \mathcal{V} \otimes \Omega_{B^\circ}^1$.

Extending $\mathcal{V}$ to all of B as a flat bundle is impractical, since the VHS degenerates along D . However, Deligne constructs a vector bundle $\bar{\mathcal{V}}$ extending $\mathcal{V}$ to D^{ls} the log-smooth part of D , along with a logarithmic connection

$$\bar{\nabla} : \bar{\mathcal{V}} \rightarrow \bar{\mathcal{V}} \otimes \Omega_{B^{\text{ls}}}^1(\log D^{\text{ls}})$$

extending ∇ . We will outline the local construction of $\bar{\mathcal{V}}$ on the smooth part of the discriminant D^{sm} ; this will be enough for the purposes of this paper, but the general case of D^{ls} is completely analogous, albeit with more variables.

Let $U \simeq \Delta^g \subseteq B^\circ$ be a polydisk which only meets $D^{\text{sm}} \subseteq D$, and let $U^\circ \simeq \Delta^* \times \Delta^{g-1}$ be the complement of the discriminant in U . Denote the restriction of $\mathcal{V}$ to U° by $\mathcal{V}_U$. The monodromy group of the restricted VHS is generated by a single element, which we will denote by $T \in \text{Aut}(V_b)$ for some $b \in U^\circ$. Let $t_1, t_2, \dots, t_g$ be the natural coordinates on U , so that U° is defined by the non-vanishing of t_1 .

By the classical Monodromy Theorem, T is quasi-unipotent, so that we can write $T = T_s \cdot T_u$, where $T_s^m = I_{2g}$ and T_u is unipotent. In particular, T admits a formal logarithm

$$N = \log T = \log T_s + \log T_u = N_s + N_u,$$

where N_u is nilpotent, uniquely defined by

$$N_u = (T_u - I_{2g}) - \frac{1}{2}(T_u - I_{2g})^2 + \dots$$

and N_s is a matrix satisfying the property

$$\exp(-2\pi i N_s) = T_s.$$

Since N_s is produced by diagonalizing T_s and applying $z \mapsto \frac{\log z}{2\pi i}$, it depends only on a choice of logarithm branch cut. As such, N_s is unique upon choosing a half-open unit interval $I \subseteq \mathbb{R}$ in which the eigenvalues of N_s lie. We construct sections for the bundle $\bar{\mathcal{V}}_U$ as follows.

Let $v = v(t_1, t_2, \dots, t_g) \in V_{\mathbb{C}}|_U \subseteq \mathcal{V}_U$ be a flat multi-section of $\mathcal{V}_U$, that is, a multi-section of the underlying $\mathbb{C}$ -local system $V_{\mathbb{C}}|_U$. We allow it to be multivalued because moving around D sends v to $T.v$. Define

$$\tilde{v}(t_1, t_2, \dots, t_g) = \exp\left(\frac{\log(t_1)N}{2\pi i}\right) \cdot v(t_1, \dots, t_g).$$

It is not hard to see that this new section is single-valued on U° , and one readily checks that $\tilde{v}$ has a removable singularity along D . For a detailed overview, refer to [Cat16].

Finally, for a frame of flat multisections $v_1, \dots, v_r \in \mathcal{V}_U$, we define the canonical extension $\overline{\mathcal{V}}_U = \bigoplus_{i=1}^r \mathcal{O}_U \tilde{v}_i$. It comes with an induced logarithmic connection

$$\nabla : \tilde{v}_j \mapsto \nabla \left(\exp \left(\frac{\log(t_1)N}{2\pi i} \right) \cdot v_j \right) = \frac{N}{2\pi i} \tilde{v}_j \otimes \frac{dt_1}{t_1}.$$

It is a highly non-trivial, but nevertheless true fact that this construction globalizes to all of B^{ls} ; that is, one glues $\overline{\mathcal{V}}_U$ along all the analytic neighborhoods U of D^{ls} the log-smooth locus of the discriminant to canonically produce an extension $\overline{\mathcal{V}}$ of $\mathcal{V}$.

Next, we turn our attention to the special case where V is a variation of Hodge structures coming from a Lagrangian fibration $f : X \rightarrow B$. In this setting, we can describe Deligne's canonical extension in another way. Generalizing a result of Matsushita, Schnell [Sch23, §4] proves that the natural isomorphism

$$\iota_\sigma : \mathcal{T}_{B^\circ} \xrightarrow{\sim} F^1 \mathcal{V}, \quad w \mapsto \sigma(w, -)$$

extends to an isomorphism

$$\iota_\sigma : \mathcal{T}_{B^{\text{ls}}} \xrightarrow{\sim} F^1 \overline{\mathcal{V}},$$

where $F^1 \overline{\mathcal{V}} := \overline{\mathcal{V}} \cap j_*(F^1 \mathcal{V})$, and $\overline{\mathcal{V}}$ is Deligne's canonical extension for which N_s has eigenvalues in $(-1, 0]$. Since the locus of D which fails to be log-smooth has codimension at least 2 in B , we may further reflexivize both sheaves to produce an isomorphism $\mathcal{T}_B \simeq (F^1 \overline{\mathcal{V}})^{\vee\vee}$; in abuse of notation, we will often drop the reflexivization on the right.

2.2. Mumford–Tate groups of weight one Hodge structures. Recall that the Deligne torus is the real algebraic group $\mathbb{S} = \text{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_{m, \mathbb{C}}$; the data of a weight one $\mathbb{Q}$ -Hodge structure is the data of a vector space $H_{\mathbb{Q}}$ together with a representation $h : \mathbb{S} \rightarrow \text{GL}(H_{\mathbb{R}})$ such that the underlying weight representation

$$\mathbb{G}_{m, \mathbb{R}} \xrightarrow{w} \mathbb{S} \xrightarrow{h} \text{GL}(H_{\mathbb{R}})$$

is defined over $\mathbb{Q}$.

Definition 2.1. The **Mumford–Tate group** of a $\mathbb{Q}$ -Hodge structure $H_{\mathbb{Q}}$ is the smallest algebraic subgroup $G \subseteq \text{GL}(H_{\mathbb{Q}})$ whose real points contain the image of h .

For a brief introduction to Mumford–Tate groups and Mumford–Tate domains, we recommend [CMSP17, Chapter 15]. However, we will summarize the necessary basic results below.

If V is a $\mathbb{Q}$ -VHS on a quasi-projective variety B° , there is a countable collection $\{Z_j\}$ of closed subvarieties of B° such that for any $b, b' \in B^\circ \setminus \bigcup_{j=1}^\infty Z_j$ we have $\mathbf{MT}(V_b) = \mathbf{MT}(V_{b'})$. For this reason, we will abuse notation and write $\mathbf{MT}(V)$ to denote the Mumford–Tate group of the Hodge structure V_b over the very general point $b \in B^\circ$.

We note that if $Z \subseteq B^\circ$ is a closed subvariety, then $\mathbf{MT}(V|_Z) \leq \mathbf{MT}(V)$. A weight one polarized VHS admits a period map to a period domain $\mathcal{P} : B^\circ \rightarrow \mathcal{A}_{g, \eta}$. Since the Mumford–Tate group only shrinks under specialization, we can make the following definition.

Definition 2.2. Let H be a weight one polarized $\mathbb{Q}$ -Hodge structure. We say that H is **Hodge-generic** if H has a maximal Mumford–Tate group in $\mathcal{A}_{g, \eta}$, that is, $\mathbf{MT}(H) = \text{GSp}_{2g, \mathbb{Q}}$.

Real group (up to isogeny)	Dimension of Mumford–Tate subdomain	Dimension of representation
$\mathrm{SL}_2(\mathbb{R})$	1	2
$\mathrm{SU}(p, n - p)$	$p(n - p)$	n
$\mathrm{SU}(1, n - 1)$	$n - 1$	$\binom{n}{c}$ $c \in \{2, 3, \dots, n - 2\}$
$\mathrm{SO}^*(2r)$	$\binom{r}{2}$	$2r$
$\mathrm{Sp}(r)$	$\binom{r+1}{2}$	$2r$
$\mathrm{SO}(2p - 2, 2)$	$2p - 2$	2^{p-1}
$\mathrm{SO}(2p - 1, 2)$	$2p - 1$	2^p

TABLE 1. Satake’s classification of real forms of irreducible weight one Mumford–Tate representations.

Notice that the VHS V naturally gives rise to two representations: the Mumford–Tate group $\mathbf{MT}(V_b)$ and the fundamental group $\pi_1(B^\circ, b)$ both act on the very general fiber V_b . One can ask if these two representations are related. A theorem of Yves André provides an affirmative answer.

Theorem 2.3 (André). *Let V be a polarized $\mathbb{Q}$ -VHS on a quasi-projective variety B° , and let $\rho : \pi_1(B^\circ, b) \rightarrow \mathrm{GL}(V_b)$ be the monodromy action on a very general fiber. Take M to be the Zariski closure of the monodromy group $\mathrm{im}(\rho) \leq \mathrm{GL}(V_b)$, and let $M^\circ \leq M$ be its identity component. Then*

$$M^\circ \trianglelefteq [\mathbf{MT}(V_b), \mathbf{MT}(V_b)].$$

On the other hand, the group $\mathbf{MT}(V_b)$ admits an action on $\mathcal{A}_{g,\eta}$. We refer to the orbit of V_b under this action as a Mumford–Tate subdomain. It can be shown that this orbit is a Hermitian locally symmetric submanifold. Consequently, the groups which can occur as $\mathbf{MT}(V_b)$ are highly constrained. Due to work of Satake [Sat67], we have a full classification of the group $\mathbf{MT}(V_b)$ and its action on V_b , up to a central isogeny. These groups are presented in Table 1, along with some useful associated invariants.

3. GENERAL SINGULAR FIBERS IN A LAGRANGIAN FIBRATION

For this section, we work with local analytic Lagrangian degenerations. To that end, let $U \simeq \Delta^g$ be a polydisk, $D \subseteq U$ an irreducible codimension one subvariety such that $U^\circ := U \setminus D \simeq \Delta^{g-1} \times \Delta^*$.

Let (X, σ) be a holomorphic symplectic manifold of dimension $2g$ equipped with a proper surjective morphism $f : X \rightarrow U$, such that X_b is Lagrangian for all $b \in U$, and X_b is smooth for $b \in U^\circ$ and singular when $b \in D$. Let X° be the preimage of U° in X . Set $V = R^1 f_* \mathbb{Z}_{X^\circ}$ to be the variation of Hodge structures underlying f . Finally, let $T \in \mathrm{Sp}_{2g}(\mathbb{Z})$ be a generator for the monodromy group of f acting on the cohomology of a general fiber X_b .

Lemma 3.1 ([BS26], Remark 6.8). *T has at most two non-trivial eigenvalues $\zeta_i, \bar{\zeta}_i$, which are the primitive i -th roots of unity for an integer i dividing six. If T has infinite order, i divides two.*

Proof. Let $N = \log T = N^{ss} + N^u$. Deligne's canonical extension extends a local frame $v_1, \dots, v_{2g} \in H^0(U^\circ, \mathcal{V})$ to a frame $\tilde{v}_1, \dots, \tilde{v}_{2g} \in H^0(U, \bar{\mathcal{V}})$, $\tilde{v}_j = \exp(\frac{zN}{2\pi i}) \cdot v_j$, for z a local coordinate on the universal cover of U° . We can choose the v_j in such a way that N has the form

$$N = \begin{pmatrix} D & S \\ 0 & D' \end{pmatrix}, \quad S = \begin{pmatrix} I_g & 0 \\ 0 & 0 \end{pmatrix}$$

This frame has the advantage that it behaves well with respect to the flat connection: if we set φ to be the local defining equation of D ,

$$\nabla \tilde{v}_k = \frac{N(\tilde{v}_k)}{2\pi i} \frac{d\varphi}{\varphi} = \frac{\alpha_k}{2\pi i} \frac{d\varphi}{\varphi} \tilde{v}_k + \frac{\beta_k}{2\pi i} \frac{d\varphi}{\varphi} \tilde{v}_{g+k}.$$

The limit mixed Hodge structures of $\mathcal{V}$ define a locally split sub-bundle $\langle \tilde{v}_1, \dots, \tilde{v}_g \rangle|_D \subseteq \bar{\mathcal{V}}|_D$. After possibly shrinking the open subset U , we may lift this to a Deligne frame $h_1, \dots, h_g, h_{g+1}, \dots, h_{2g} \in H^0(U, \bar{\mathcal{V}})$,

$$F^1 \bar{\mathcal{V}} = \langle h_1, \dots, h_g \rangle \subseteq \bar{\mathcal{V}},$$

agreeing with $\langle \tilde{v}_1, \dots, \tilde{v}_g \rangle$ on D .

Now we have two exact sequences

$$\begin{array}{ccccc} \langle \tilde{v}_{g+1}, \dots, \tilde{v}_{2g} \rangle & \longrightarrow & \bar{\mathcal{V}} & \longrightarrow & \bar{\mathcal{V}} / \langle \tilde{v}_{g+1}, \dots, \tilde{v}_{2g} \rangle \\ \parallel & & \parallel & & \downarrow g \\ \langle \tilde{v}_{g+1}, \dots, \tilde{v}_{2g} \rangle & \longrightarrow & \bar{\mathcal{V}} & \longrightarrow & \bar{\mathcal{V}} / \langle h_{g+1}, \dots, h_{2g} \rangle \end{array}$$

Thus the two splittings differ by an element of $\mathrm{Hom}(\bar{\mathcal{V}} / \langle \tilde{v}_{g+1}, \dots, \tilde{v}_{2g} \rangle, \langle \tilde{v}_{g+1}, \dots, \tilde{v}_{2g} \rangle)$. In other words, the isomorphism g has the form

$$h_j = \tilde{v}_j + \sum_{k=g+1}^{2g} g_{jk} \tilde{v}_k, \quad j = 1, \dots, g.$$

Recall from [BS26] and our summary above the isomorphism $\mathcal{T}_U \simeq F^1\overline{\mathcal{V}} = \langle h_1, \dots, h_g \rangle$. This isomorphism is compatible with Lie bracket, so we have

$$\begin{aligned}
[h_j, h_k] &= \nabla_{h_j} \tilde{v}_k - \nabla_{h_k} \tilde{v}_j + \sum_{\ell=g+1}^{2g} h_j(g_{k\ell}) \tilde{v}_\ell + g_{k\ell} \nabla_{h_j} \tilde{v}_\ell - h_k(g_{j\ell}) \tilde{v}_\ell - g_{j\ell} \nabla_{h_k} \tilde{v}_\ell \\
&= \frac{\alpha_k}{2\pi i} \frac{d\varphi(h_j)}{\varphi} \tilde{v}_k - \frac{\alpha_j}{2\pi i} \frac{d\varphi(h_k)}{\varphi} \tilde{v}_j + \frac{\beta_k}{2\pi i} \frac{d\varphi(h_j)}{\varphi} \tilde{v}_{g+k} - \frac{\beta_j}{2\pi i} \frac{d\varphi(h_k)}{\varphi} \tilde{v}_{g+j} + \\
&\quad + \sum_{\ell=g+1}^{2g} \left(h_j(g_{k\ell}) - h_k(g_{j\ell}) + \frac{\alpha_\ell}{2\pi i} \left(g_{k\ell} \frac{d\varphi(h_j)}{\varphi} - g_{j\ell} \frac{d\varphi(h_k)}{\varphi} \right) \right) \tilde{v}_\ell \\
&= \frac{\alpha_k}{2\pi i} \frac{d\varphi(h_j)}{\varphi} h_k - \frac{\alpha_j}{2\pi i} \frac{d\varphi(h_k)}{\varphi} h_j + \frac{\beta_k}{2\pi i} \frac{d\varphi(h_j)}{\varphi} \tilde{v}_{g+k} - \frac{\beta_j}{2\pi i} \frac{d\varphi(h_j)}{\varphi} \tilde{v}_{g+j} + \\
&\quad + \sum_{\ell=g+1}^{2g} \left(g_{j\ell} - g_{k\ell} + h_j(g_{k\ell}) - h_k(g_{j\ell}) + \frac{\alpha_\ell}{2\pi i} \left(g_{k\ell} \frac{d\varphi(h_j)}{\varphi} - g_{j\ell} \frac{d\varphi(h_k)}{\varphi} \right) \right) \tilde{v}_\ell \\
&= \frac{\alpha_k}{2\pi i} \frac{d\varphi(h_j)}{\varphi} h_k - \frac{\alpha_j}{2\pi i} \frac{d\varphi(h_k)}{\varphi} h_j.
\end{aligned}$$

The last equality follows from the fact that the $\tilde{v}_\ell$ are not contained in $\mathcal{T}_U$ when $\ell \geq g+1$.

Now, for $[h_j, h_k]$ to be holomorphic, we need either $\alpha_k = 0$ or φ divides $d\varphi(h_j)$. The latter is equivalent to h_j being tangent to D . But D is of codimension one, so at least one of the h_j is not tangent to D , say h_λ . This forces $\alpha_k = 0$ for all $k \neq \lambda$.

Notice additionally that g_{jk} all vanish on D . Thus, when restricting to D , the higher $\tilde{v}_\ell$ terms in the expression above yield

$$0 = \frac{\beta_k}{2\pi i} \frac{d\varphi(h_j)}{\varphi} \tilde{v}_{g+k} - \frac{\beta_j}{2\pi i} \frac{d\varphi(h_k)}{\varphi} \tilde{v}_{g+j} + \sum_{\ell=g+1}^{2g} (h_j(g_{k\ell}) - h_k(g_{j\ell})) \tilde{v}_\ell$$

The sum on the right is holomorphic along D , and φ does not divide $d\varphi(h_\lambda)$. Hence $\beta_k = 0$ for all $k \neq \lambda$. Hence we may lose no generality in writing that $\lambda = 1$; that is, the only possibly non-trivial eigenvalues are α_1 and the only possibly non-semisimple eigenvector is $\tilde{v}_1$.

Observe that T is now an integer matrix with at most two non-trivial eigenvalues. By Grothendieck, all eigenvalues of T are roots of unity; if ζ_i is a primitive i -th root of unity, all other primitive i -th roots of unity are in its Galois orbit, and thus also occur as eigenvalues of T . Thus if ϕ is Euler's totient function, we must have $\phi(i) \leq 2$. This forces $i = 1, 2, 3, 4$, or 6 . Furthermore, when T has infinite order, there is only one non-trivial Jordan block, forcing $\zeta_i = \bar{\zeta}_i$, or equivalently $\zeta_i = \pm 1$. $\square$

4. CLASSIFICATION OF LAGRANGIAN MUMFORD–TATE GROUPS

4.1. Existence of infinite local monodromy. Now we work in the compact setting. Let X be a primitive symplectic variety of dimension $2g$ which admits a maximally varying Lagrangian fibration $f : X \rightarrow B$. Take V to be the underlying variation of Hodge structures on B° .

Proposition 4.1. *There exists an analytic neighborhood $U \simeq \Delta^g \subseteq B$ meeting the discriminant $D \subseteq B$ in a codimension one subvariety $D_U := D \cap U \subseteq U$, such that the local monodromy of f around D_U is infinite.*

Before proving this proposition, we remark on an important consequence. For one, recall from Hwang–Oguiso [HO09] that $D \subseteq B$ is either empty or codimension one. In the case where X is primitive symplectic, we can rule out the former case entirely.

Corollary 4.2. *Let X be a primitive symplectic variety, and let $f : X \rightarrow B$ be a Lagrangian fibration. Then the discriminant locus $D \subseteq B$ of f has codimension one.*

Proof. Since X is primitive symplectic, f is either maximally varying or isotrivial. By Theorem 4.1, the discriminant of f has a codimension one component whenever f varies maximally, so we may take f to be isotrivial. Now if $D = \emptyset$, f is the trivial fibration and $X \simeq X_b \times B$. But by Kunneth, we must now have $H^1(X, \mathbb{C}) \neq 0$, a contradiction. $\square$

Proof of Theorem 4.1. Our proof will use methods of [GMMT25] to analyze the period image of V . Since $V_{\mathbb{Q}}$ is irreducible, the period image lands in an indecomposable special subvariety of $\mathcal{A}_{g,\eta}$. By [GMMT25, Theorem 4.5], the g -dimensional period image is not Hodge-generic inside of any special indecomposable non-compact subvariety. Thus we only need to show that the period image is not contained in a compact special subvariety.

Let $G \subseteq [\mathbf{MT}(V_{\mathbb{Q}}), \mathbf{MT}(V_{\mathbb{Q}})]$ be the derived Mumford–Tate group of V . By [BT65, Corollary 8.5], an algebraic group over a number field is anisotropic (contains no positive-dimensional split tori) if and only if it contains no non-trivial unipotent elements. By Schmid’s nilpotent orbit theorem, for G this is equivalent to defining a compact special subvariety in $\mathcal{A}_{g,\eta}$.

Note that G is anisotropic if and only if its $\mathbb{Q}$ -simple factors are, since a unipotent element in a factor induces a unipotent element in G and vice versa. Hence we may assume without loss of generality that G is a $\mathbb{Q}$ -simple algebraic group. By the classical theorem [GMMT25, Corollary 2.7], we can write $G = \text{Res}_{K/\mathbb{Q}} H$, for K a totally real field and H a geometrically simple K -algebraic group. But again, G is anisotropic if and only if H is anisotropic.

We will prove that H is not anisotropic. This will show that the image of the period map is not contained in a compact special subvariety. To do this, we will apply the results of [Sch85]. Let $U \subseteq V_K$ the irreducible K -representation induced by the factor $H \leq G_K$.

Lemma 4.3 ([GMMT25], §5.1). *Suppose that $H = \text{Aut}(U, \eta)^{\text{ad}}$ is non-compact at all real places. Assume that one of the following conditions holds:*

- (q) U is a K -vector space, η is symmetric bilinear, and $\dim_K U \geq 5$.
- (uF) U is an L -vector space for L/K a quadratic extension with involution $*$, η is a $*$ -hermitian form, and $\dim_L U \geq 3$.
- (uH) U is a right $\mathbb{H}$ -module for $\mathbb{H}/K$ a quaternion division algebra with involution $*$, η is a $*$ -hermitian form, and $\dim_{\mathbb{H}} U \geq 2$.
- (uD) U is a right $\mathbb{D}$ -module, for $\mathbb{D}$ a non-commutative division algebra over L a quadratic extension of K with involution $*$, η is a $*$ -hermitian form, and $\dim_{\mathbb{D}} U \geq 3$.
- (skH) U is a right $\mathbb{H}$ -module, for $\mathbb{H}$ a quaternion division algebra over K with canonical involution $*$, η is a skew-hermitian form, and $\dim_{\mathbb{H}} U \geq 4$.

Then H is anisotropic.

In our setting, V varies maximally, so H must be non-compact at all real places. Thus we may rule out all cases one-by-one. We follow the exact analysis of [GMMT25, §7.2]. Satake

classified all possible real forms of H up to isogeny, and we analyze each case. Let S denote the special subvariety associated to G .

H has type A_1 : We must have $H = \mathrm{Nm}(\mathbb{H})$ and $\dim U = 2$ for all i . Write $\dim V = 2g = k \dim U = 2kh$. If $k = 1$, $H = \mathrm{SL}_{2,K}$ is not anisotropic. Otherwise, $g \leq \dim S = \frac{g}{h}$ yields a contradiction.

H has type D_4 : $U_{\mathbb{C}}$ is a standard orthogonal representation of dimension $h = 8$. Since U is not symplectic, its dual must also occur in the decomposition of V , so $\dim V = 16k$, while $\dim S = 6$. Thus $g = 8k \leq 6$, a contradiction.

H has type I : We are in (uF) and (uD) of Theorem 4.3, so $H = \mathrm{SU}(\mathbb{D}^r, \eta)$ with $r \leq 2$ and $\dim_K U \leq 4$. Here $\mathbb{D}$ has degree $\delta \geq 2$ over a CM extension L/K . We may identify $U_{\mathbb{C}} \simeq M_{\delta \times \delta}(\mathbb{C})^{\oplus r}$ as $H(\mathbb{C})$ -representations. Moreover, U is not self-dual, so its dual occurs in the decomposition of $V_{\mathbb{C}}$. In particular,

$$\dim V = 2g = 2k \dim U = 2k\delta^2 r.$$

On the other hand, $\dim S$ is maximized when the signature of $H(\mathbb{C})$ is $(\frac{r\delta}{2}, \frac{r\delta}{2})$. Even in this case,

$$g = k\delta^2 r \leq \dim S \leq k \cdot \frac{\delta r}{2} \cdot \frac{\delta r}{2} = \frac{1}{4}k\delta^2 r^2,$$

so that $r \geq 4$, a contradiction.

H has type I' : H is the same as in type I , and $U \simeq \bigwedge^c \mathbb{C}^{r\delta}$. Now $\dim U = \binom{r\delta}{c}$, while $\dim S = r\delta - 1$. Since U is not self-dual, we cannot have $g \leq \dim S$ even when $c = 1$.

H has type II : Here $G = \mathrm{SU}(\mathbb{H}^r, \eta)$ with $r \geq 2$ and η skew-hermitian. We are in the case (skH) of Theorem 4.3, so $2 \leq \dim_{\mathbb{H}} U \leq 3$. Note again that U is not self-dual, so that its dual also occurs in the decomposition of $V_{\mathbb{C}}$. Thus $\dim V = 2g = 2k \dim U = 4kr$. On the other hand, $\dim S = k \frac{r(r-1)}{2}$. Thus $g = 2kr \leq k \frac{r(r-1)}{2}$, so that $4 \leq r - 1$, a contradiction.

H has type III : There are two possibilities here. Either $H = \mathrm{Sp}(K^{2r}, \eta)$, or $H = \mathrm{SU}(\mathbb{H}^r, \eta')$. The former case is always anisotropic, so we need not consider it. When $r \geq 2$, we are in case (uH) of Theorem 4.3, so S is non-compact. When $r = 1$, H has type A_1 , which we have already disproved.

H has type IV : H must be a form of $\mathrm{SO}_n^{\mathrm{ad}}$, for $n \geq 5$. This puts us into the cases (q) and (skH) of Theorem 4.3. In either case, H is anisotropic.

□

4.2. Splitting the monodromy group also splits the representation. Let S be the special Mumford–Tate group of $V_{\mathbb{Q}}$, and let $\Lambda = \mathrm{im}(\rho : \pi_1(B^{\circ}, b) \rightarrow \mathrm{Sp}_{2g, \mathbb{Q}})$. Define its Zariski closure $\mathrm{Mon}(V_{\mathbb{Q}}) = \bar{\Lambda}$; it admits an exact sequence

$$1 \rightarrow \mathrm{Mon}(V_{\mathbb{Q}})^{\circ} \rightarrow \mathrm{Mon}(V_{\mathbb{Q}}) \rightarrow \Gamma \rightarrow 1,$$

where Γ is the component group of $\mathrm{Mon}(V_{\mathbb{Q}})$. By André, $\mathrm{Mon}(V_{\mathbb{Q}})^{\circ}$ is a normal subgroup of S .

Lemma 4.4. *There exists a branched Galois cover of normal projective varieties $\pi : B' \rightarrow B$ such that $\mathrm{Mon}(\pi^* V_{\mathbb{Q}}) = \mathrm{Mon}(V_{\mathbb{Q}})^{\circ}$.*

Proof. Construct $\pi : B'^{\circ} \rightarrow B^{\circ}$ as follows: let Λ° be the preimage of $\mathrm{Mon}(V_{\mathbb{Q}})^{\circ}$ under the natural inclusion. It is the preimage of a normal subgroup, and is therefore normal in Λ . We

have an exact sequence

$$1 \rightarrow \Lambda^\circ \rightarrow \Lambda \rightarrow \Gamma \rightarrow 1.$$

Define B'° to be the étale cover corresponding to the surjection $\Lambda \rightarrow \Gamma$; that is, $\Lambda^\circ = \pi_1(B'^\circ, b')$. Because Λ° is normal in Λ , the cover is Galois. Now we may define B'° to be the normalization of B'° in the function field $K(B)$. $\square$

To understand the monodromy of $\pi^*V_{\mathbb{Q}}$, we will use André’s theorem, which implies that $\text{Mon}(V_{\mathbb{Q}})^\circ \trianglelefteq S$. The remainder of this section will be dedicated to understanding the action of S on $V_{\mathbb{Q}}$.

Proposition 4.5. *There exists an isomorphism of $\mathbb{C}$ -Hodge structures $\pi^*V_{\mathbb{C}} = U_1 \oplus \cdots \oplus U_k$, where U_i has the compatible structure of an irreducible $\mathfrak{h}_i$ -representation and $\mathfrak{h}_j$ acts trivially on U_i for $j \neq i$.*

Proof. We have an isomorphism of complex Lie algebras $\mathfrak{s}_{\mathbb{C}} = \mathfrak{h}_1 \oplus \cdots \oplus \mathfrak{h}_k$ coming from the isogeny decomposition. In particular, the decomposed Lie algebra acts on $V_{\mathbb{C}}$ by

$$d\rho : \mathfrak{h}_1 \oplus \cdots \oplus \mathfrak{h}_k \rightarrow \text{Aut}(V_{\mathbb{C}}).$$

Let U be an irreducible factor of $\pi^*V_{\mathbb{C}}$; we can write it as a tensor product $U = W_1 \otimes \cdots \otimes W_k$, where W_i is an irreducible $\mathfrak{h}_i$ -representation.

Let $\mu : \mathbb{G}_{m, \mathbb{C}} \rightarrow S(\mathbb{C})$ be the Hodge cocharacter. Since $V_{\mathbb{Q}}$ is weight 1, $\mu(z)$ acts by z or 0. Consequently, $d\mu : \mathbb{C} \rightarrow \mathfrak{s}_{\mathbb{C}} \rightarrow \text{Aut}(V_{\mathbb{C}})$ acts by matrices with eigenvalues in $\{0, 1\}$. Denote by $\mu_i : \mathbb{C} \rightarrow \mathfrak{h}_i \rightarrow \text{Aut}(W_i)$ the projection to the i -th factor. Suppose for $z \in \mathbb{C}$ and $w_i \in W_i$ we have $d\mu(z).w_i = w_i$; then

$$d\mu(z).(w_1 \otimes \cdots \otimes w_k) = (\lambda_1 + \cdots + \lambda_k).(w_1 \otimes \cdots \otimes w_k),$$

where λ_j are the eigenvalues of $d\mu(z)$ acting on W_j . Since $\lambda_i = 1$ and $\lambda_1 + \cdots + \lambda_k \in \{0, 1\}$, we must have $\lambda_j = 0$ for all $j \neq i$.

Lemma 4.6. *$\mathfrak{h}_j$ acts trivially on W_j for $j \neq i$.*

Proof. S is the smallest $\mathbb{Q}$ -algebraic group whose real points contain the image of the Hodge cocharacter. As such, if $X \in \mathfrak{s}_{\mathbb{C}}$ is a generator of the image of $d\mu$, its projection to $\mathfrak{h}_j$ must be $X_j \neq 0$ for all $j = 1, \dots, k$. Furthermore, $\mathfrak{h}_j$ is simple for all j , so that $[\mathfrak{h}_j, \mathfrak{h}_j] = \mathfrak{h}_j$. In particular, X_j is traceless on W_j .

Now if W_j is non-trivial, X_j must have at least one positive integer eigenvalue and at least one negative integer eigenvalue. Let $\lambda_j \geq 1$ be the largest eigenvalue of X_j acting on W_j . By the same argument, $\lambda_i \geq 1$. This forces X to have an eigenvalue of at least two on U , a contradiction. Thus the image of the Deligne torus X_j acts trivially on W_j for $j \neq i$.

Note that we have not yet shown that $\mathfrak{h}_j$ acts trivially, only that the representation W_j has level zero. Indeed, one may consider Mumford’s famous example of a non-trivial family of Abelian 4-folds with Mumford–Tate group isogenous to $\text{SL}_2(\mathbb{R}) \times \text{SU}(2)^2$ [Mum69]. To rule out such examples, we will need to leverage some actual geometry native to Lagrangian fibrations.

Let $D_1 \subseteq B$ be a component of the discriminant with infinite local monodromy; such a component exists by Theorem 4.1. Let T be a local monodromy operator for this component; its logarithm N belongs to $\mathfrak{h}_1 \oplus \cdots \oplus \mathfrak{h}_k$. Since $\mathfrak{h}_i$ acts with level one on W_i , we may choose T such that the projection $N_i \in \mathfrak{h}_i$ acts non-trivially. By Theorem 3.1, N has rank one, so that all other projections $N_j = 0$, for $j \neq i$.

Now observe that every other W_j is forced to be one-dimensional. Indeed, suppose $\dim W_1 \geq 2$; then for every basis vector $w_{1,\alpha} \in W_1$,

$$N.(w_{1,\alpha} \otimes \cdots \otimes w_i \otimes \cdots \otimes w_k) = w_{1,\alpha} \otimes \cdots \otimes N_i.w_i \otimes \cdots \otimes w_k$$

provides a distinct linearly independent vector in $\text{im}(N)$, a contradiction.

Finally, since the action of $\mathfrak{h}_j$ on W_j is forced to be given by characters, it factors through the Abelianization $\mathfrak{h}_j/[\mathfrak{h}_j, \mathfrak{h}_j]$. But $\mathfrak{h}_j$ is simple, so that this is the trivial group. Hence only $\mathfrak{h}_i$ acts non-trivially on $W_1 \otimes \cdots \otimes W_k$, as needed. $\square$

Thus we may write $\pi^*V_{\mathbb{C}} = (B_1 \otimes U_1) \oplus \cdots \oplus (B_k \otimes U_k)$; however, this is only a decomposition of infinitesimal representations. Next, we will show that the $B_i \otimes U_i$ are in fact representations of $S(\mathbb{C})$. To that end, write

$$\mathfrak{k}_i = \mathfrak{h}_1 \oplus \cdots \oplus \widehat{\mathfrak{h}_i} \oplus \cdots \oplus \mathfrak{h}_k;$$

this allows us to describe $B_i \otimes U_i$ as $\text{Ann}(\mathfrak{k}_i)$. We now claim that for all $g \in S(\mathbb{C})$, $gB_i \otimes U_i = B_i \otimes U_i$. To see this, choose any $X \in \mathfrak{k}_i$ and $u_i \in B_i \otimes U_i$; by Lie theory,

$$0 = g.(X.u_i) = g \frac{d}{dt} \Big|_{t=0} e^{tX} g^{-1} g.u_i = (\text{Ad}(g)X).(g.u_i).$$

But Ad is a continuous action of $S(\mathbb{C})$ a connected group on the discrete set of simple ideals of $\mathfrak{s}_{\mathbb{C}}$. Hence the action does not permute the simple ideals, so that $\text{Ad}(g)\mathfrak{h}_i = \mathfrak{h}_i$, and analogously for $\mathfrak{k}_i$. This implies that $\text{Ad}(g)X \in \mathfrak{k}_i$, so that $g.u_i \in \text{Ann}(\mathfrak{k}_i) = B_i \otimes U_i$, as needed.

Now we claim that $\mathfrak{h}_1 = \cdots = \mathfrak{h}_k$. Indeed, by [Var26], $V_{\mathbb{C}}$ is irreducible, so by Clifford's theorem [Cli37], Γ permutes the $B_i \otimes U_i$ factors transitively; that is, $(B_i \otimes U_i)^\gamma = B_j \otimes U_j$, for all $\gamma \in \Gamma$. But recall that $U_i = \text{Ann}(\mathfrak{k}_i)$, so conjugation by γ induces an outer isomorphism $\mathfrak{k}_i = \mathfrak{k}_j$, forcing $\mathfrak{h}_i = \mathfrak{h}_j$, as needed.

Finally, we claim that $B_i = \mathbb{C}$ for all i . Indeed, $V_{\mathbb{C}}$ varies maximally, so $\pi^*V_{\mathbb{C}}$ also varies maximally, meaning that no power of a Hodge structure shows up in the very general fiber. This shows that $\pi^*V_{\mathbb{C}} = U_1 \oplus \cdots \oplus U_k$, as needed. $\square$

Notice that $S(\mathbb{C})$ is still merely isogenous to a power of a simple group. We lose no generality in taking H^k to be the universal cover of $S(\mathbb{C})$; thus $\pi^*V_{\mathbb{C}} = U_1 \oplus \cdots \oplus U_k$ acquires the structure of a H^k -representation. Furthermore, the i -th factor H_i acts non-trivially on U_i and trivially on U_j for $j \neq i$.

Proposition 4.7. *Let $G = \mathbf{MT}(V_{\mathbb{Q}})$ be the full Mumford–Tate group, and $\tilde{G}$ its universal cover, $\tilde{G} = Z \times H_1 \times \cdots \times H_k$. There exists an intermediate extension $G' \twoheadrightarrow G$ for which the Deligne torus $h : \mathbb{S} \rightarrow G$ lifts to $h' : \mathbb{S} \rightarrow G'$. Furthermore, we may write G' as a diagonal quotient*

$$G' = \tilde{G}/(\underline{\zeta}) = \frac{Z \times H_1 \times \cdots \times H_k}{(\zeta_0, \zeta_1, \dots, \zeta_k)}$$

for $\zeta_0 \in Z$ and $\zeta_i \in Z(H_i)$.

Proof. In order for the Deligne torus to lift to an intermediate cover $G' \rightarrow G$, we must have $h' : \mathbb{S} \rightarrow G'$ acting by a torus representation on U_i . In other words, $dh' = X_0 + X_1 + \cdots + X_k$ must have integer eigenvalues. This is the same as requiring $\exp(2\pi i \cdot dh') = 1 \in G'$. Thus we define $\zeta_i = \exp(2\pi i X_i)$, $i = 0, \dots, k$, and take $\underline{\zeta} = (\zeta_0, \zeta_1, \dots, \zeta_k) \in \tilde{G}(\mathbb{C})$. By construction, the Deligne torus will lift to $G' = \tilde{G}/\underline{\zeta}$. $\square$

To better understand why we generally cannot lift the Deligne torus to the universal cover $\tilde{G}$, consider the following example.

Example 4.8. Suppose $G = \mathrm{GL}_2 \times_{\mathbb{G}_m} \mathrm{GL}_2$, acting on $U_1 \oplus U_2$ by the standard representation. Then $\tilde{G}(\mathbb{C}) = \mathbb{G}_m(\mathbb{C}) \times \mathrm{SL}_2(\mathbb{C})^2$. We can decompose the Hodge co-character as

$$\mu = \begin{pmatrix} z & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & z & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad d\mu = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = X_0 + X_1 + X_2.$$

Write the trace $\mathrm{Tr}(X|_{U_1}) = \mathrm{Tr}((X_0 + X_1)|_{U_1}) = \mathrm{Tr}(X_0|_{U_1}) = 1$. But X_0 is central, so it is diagonal and acts by multiplication by $1/2$. Consequently,

$$X_0 = \begin{pmatrix} 1/2 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 \\ 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 1/2 \end{pmatrix}, \quad X_1 = \begin{pmatrix} 1/2 & 0 & 0 & 0 \\ 0 & -1/2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad X_2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & -1/2 \end{pmatrix}.$$

Now if we exponentiate these generators, we get

$$\zeta_0 = -1 \in \mathbb{G}_m, \quad \zeta_1 = \zeta_2 = -I_2 \in Z(\mathrm{SL}_2).$$

Notice that $G' = \tilde{G}/(\underline{\zeta}) = G$, so the Deligne torus lifts no further than G . $\square$

Now take the derived subgroup of G' :

$$[G', G'] = [\tilde{G}/N, \tilde{G}/N] = \frac{[\tilde{G}, \tilde{G}]N}{N} = \frac{[\tilde{G}, \tilde{G}]}{[\tilde{G}, \tilde{G}] \cap N},$$

where $N = \langle \underline{\zeta} \rangle$. Notice that all the ζ_i are central, so that the intersection $[\tilde{G}, \tilde{G}] \cap N$ is trivial. Since $[\tilde{G}, \tilde{G}] = H_1 \times \cdots \times H_k$, we get the following corollary.

Corollary 4.9. *Let $U'_1 \oplus \cdots \oplus U'_k$ be the induced representation of G' on $\pi^*V_{\mathbb{C}} = U_1 \oplus \cdots \oplus U_k$. The derived subgroup $[G', G'] = H_1 \times \cdots \times H_k$ is a power of a simple group, and acts on $U'_1 \oplus \cdots \oplus U'_k$ with each H_i acting non-trivially on U'_i and trivially on U'_j for $j \neq i$.*

4.3. Proof of the main theorem. Now we are ready to prove the main theorem.

Theorem 4.10. *Let X be a primitive symplectic variety of dimension $2g$, and let $f : X \rightarrow B$ be a Lagrangian fibration. Let G be the derived Mumford–Tate group of f .*

$$G = \begin{cases} (\mathrm{Sp}_{2g/k})^k & : f \text{ varies maximally,} \\ 1 & : f \text{ is isotrivial with fiber isogenous to } E^g \\ & \text{for } E \text{ a CM elliptic curve,} \\ \mathrm{SL}_2 & : \text{otherwise.} \end{cases}$$

Proof. We first deal with the latter two cases. By [Bak25], if f does not vary maximally, it is isotrivial, so we can refer to [KLM26]. Namely, the fiber of f is isogenous to E^n , for E an elliptic curve. Now G is SL_2 unless E is CM, in which case G is trivial. Thus we may assume that f varies maximally.

As before, take $\pi : B' \rightarrow B$ be the cover over which $\pi^*V_{\mathbb{Q}} = U_1 \oplus \cdots \oplus U_k$ splits, and let Γ be the automorphism group of the cover, acting transitively on the U_i . We claim that each

of the U_i is an isotypic factor; that is, $U_i \not\cong U_j$ for $i \neq j$. To see that this holds, suppose that $U_1 \cong U_2$. By Theorem 4.1, there exists a component D_1 of the discriminant which has non-finite local monodromy. Without loss of generality, suppose that U_1 degenerates along D_1 . Then U_2 also degenerates along D_1 . But by Theorem 3.1, every general degeneration is of nilpotent rank one, a contradiction.

Hence we must have $U_i \not\cong U_j$ for $i \neq j$. We now claim that the splitting is defined over $\mathbb{Q}$. Indeed, the splitting occurs over some number field K , so we have a $\mathbb{Q}$ -irreducible factor $W \subseteq V_{\mathbb{Q}}$ which splits as $W_K = U_1 \oplus \cdots \oplus U_{\ell}$. But now the Galois group $\text{Gal}(K/\mathbb{Q})$ transitively permutes the $U_1, \dots, U_{\ell}$. Now as above, suppose U_1 degenerates over $D_1 \subseteq D$. The limit variation of mixed Hodge structures of V over D_1 is defined over $\mathbb{Q}$, while that of U_1 is not; thus any element of the Galois orbit of U_1 must degenerate over D_1 like U_1 . But by Theorem 3.1, at most one of the U_i can degenerate, forcing $\ell = 1$, and therefore $K = \mathbb{Q}$.

Now any irreducible factor of G is defined over $\mathbb{Q}$. Set $U = U_1$ and $H = H_1$, and consider the action of H on U . The group H is geometrically simple, so that $H(\mathbb{C})$ is one of $\text{SL}_{2h}(\mathbb{C})$, $\text{Sp}_{2h}(\mathbb{C})$, or $\text{SO}_{2h}(\mathbb{C})$. However, we may rule out $\text{SL}_{2h}(\mathbb{C})$ and $\text{SO}_{2h}(\mathbb{C})$, since U comes from a family of Abelian varieties of dimension h .

Recall that $\text{Sp}_{2h}(\mathbb{C})$ has no non-trivial outer automorphisms. Thus H is an inner form of Sp_{2h} . Since U comes from a family of Abelian varieties (up to isogeny), $H(\mathbb{R}) = \text{Sp}_{2h}(\mathbb{R})$; the only alternative would be $\text{Sp}(p, q)$, which is non-Hermitian when $q \neq 0$, for which U is isotrivial. Now H must be of the form $\text{Sp}_r(\mathbf{D})$, for $\mathbf{D}$ a division algebra with center $\mathbb{Q}$, so that the fibers of U have the form $U_b = \mathbf{D}^r$. Maximal parabolic subgroups of H are in bijection with totally isotropic $\mathbf{D}$ -subspaces $W \subseteq U_b$ via orbit-stabilizer. In particular, since $\dim_{\mathbb{Q}} \mathbf{D} = d > 1$, the rank of the local monodromy logarithm of U around D_1 is at least d , contradicting Theorem 3.1. Therefore $H = \text{Sp}_{2h}$, and $G = \text{Sp}_{2h}^k$, concluding the proof. $\square$

Note that this allows us to compute the center of the special Mumford–Tate group. Since special endomorphisms are ruled out, the only endomorphisms which can commute with the special Mumford–Tate group are rational scalars. If we take c to be such a scalar, it satisfies

$$c^2 \psi(v, w) = \psi(cI_{2g}.v, cI_{2g}.w) = \psi(v, w),$$

with respect to the polarization ψ , forcing $c^2 = 1$ and $c = \pm 1$. However, since the special Mumford–Tate group is generated by the derived Mumford–Tate group and the identity component of the center, it is forced to be equal to G . This allows us to easily compute the actual Mumford–Tate group.

Corollary 4.11. *Let X be a primitive symplectic variety of dimension $2g$, and let $f : X \rightarrow B$ be a Lagrangian fibration. The Mumford–Tate group of f has the form*

$$\mathbf{MT}(X_b) = \begin{cases} (\mathbb{G}_m \times \text{Sp}_{2g/k}^k)/\{\pm 1\} & : f \text{ varies maximally,} \\ \mathbb{G}_m & : f \text{ is isotrivial with fiber isogenous to } E^g \\ & \text{for } E \text{ a CM elliptic curve,} \\ \text{GL}_2 & : \text{otherwise.} \end{cases}$$

4.4. An application to a question of Li–Tosatti. We will first need to make our notation a bit more precise. When we refer to “the” special Kähler metric ω_{SK} , we will simply fix one, since they are unique up to scaling. For any sufficiently small analytic neighborhood $U \subseteq B^\circ$ there is a local coordinate system $z_1, \dots, z_g$ such that the entries of the period

matrix $\mathbf{T}$ satisfy the Donagi–Markman symmetry condition [DM94]:

$$(4.1) \quad \frac{\partial \tau_{ij}}{\partial z_k} = \frac{\partial \tau_{ik}}{\partial z_j}, \quad 1 \leq i, j, k \leq g.$$

Proposition 4.12. *Let X be primitive symplectic, and $f : X \rightarrow B$ be a Lagrangian fibration with period map $\mathcal{P} : B^\circ \rightarrow \mathcal{A}_{g,\eta}$ and period matrix $\mathbf{T}$, written as*

$$\mathcal{P}(b) = \begin{pmatrix} \mathbf{T}(b) \\ I_g \end{pmatrix}, \quad \mathbf{T} = \begin{pmatrix} \tau_{11} & \cdots & \tau_{1g} \\ \vdots & \ddots & \vdots \\ \tau_{1g} & \cdots & \tau_{gg} \end{pmatrix}$$

Suppose that $\mathbf{T}$ has a constant diagonal entry τ_{ii} in some analytic neighborhood $U \subseteq B^\circ$. Then f is isotrivial.

Proof. Without loss of generality, take $i = 1$. By directly applying (4.1), the condition that τ_{11} is constant in all directions implies that

$$\frac{\partial \tau_{11}}{\partial z_1} = \frac{\partial \tau_{12}}{\partial z_1} = \cdots = \frac{\partial \tau_{1k}}{\partial z_1} = 0.$$

Let V be the $\mathbb{Z}$ -VHS underlying f on the locus of smooth fibers B° . Choose a very general point $p \in B^\circ$, and let C_p be a (locally defined) complex curve given by integrating $\partial/\partial z_1$ through p . We get a real Hodge substructure

$$(4.2) \quad E = \begin{pmatrix} \tau_{11} \\ \tau_{21} \\ \vdots \\ \tau_{g1} \\ | \\ e_1 \\ | \end{pmatrix} \oplus \begin{pmatrix} \bar{\tau}_{11} \\ \bar{\tau}_{21} \\ \vdots \\ \bar{\tau}_{g1} \\ | \\ e_1 \\ | \end{pmatrix} \subseteq V_{\mathbb{R},p},$$

which is preserved by moving along C_p . In other words, E is contained in the fixed part of $V_{\mathbb{R}}|_{C_p}$.

We can interpret C_p more moduli-theoretically. Recall that we have $\mathcal{P} : B^\circ \rightarrow \mathcal{A}_g$; let $\tilde{\mathcal{P}} : U \rightarrow \mathbb{H}_g$ be a local lift of $\mathcal{P}$ to the universal cover of $\mathcal{A}_{g,\eta}$. Then $\tilde{\mathcal{P}}(U)$ meets the locus $\{E\} \times \mathbb{H}_{g-1}$ along the curve $\tilde{\mathcal{P}}(C_p)$. Observe that this is an atypical intersection: the image of $\tilde{\mathcal{P}}$ is g -dimensional, while the codimension of $\{E\} \times \mathbb{H}_{g-1}$ is also $\binom{g+1}{2} - \binom{g}{2} = g$, so the expected dimension of the intersection is $g - g = 0$. As a result, we can apply the functional transcendence result of Bakker–Tsimerman [BT25, Theorem 1.3]: because $\tilde{\mathcal{P}}(U) \cap (\mathbb{H}_{g-1} \times \{E\})$ is atypical, $\mathcal{P}(C_p)$ is contained in a weakly special subvariety of $\mathcal{A}_g$.

However, recall that p was a very general point. Hence there is a weakly special subvariety $Z \subsetneq \mathcal{A}_{g,\eta}$ containing $\mathcal{P}(B^\circ)$. Such a subvariety by definition arises from a normal subgroup H of the Mumford–Tate group $G = \mathbf{MT}(Z)$ of the very general point of Z . Now recall Theorem 4.11; we want to rule out the case where G is isogenous to $(\mathrm{Sp}_{2h})^{g/h}$, for some h dividing g .

If $h = g$, then we are immediately done, since $G = \mathrm{GSp}_{2g}$ has only $H = \mathbb{G}_m$ as a non-trivial normal subgroup, forcing $\mathcal{P}(C_p)$ to be a single point. This implies that f does not vary maximally, so it is isotrivial. On the other hand, if G is isogenous to $(\mathrm{Sp}_{2h})^{g/h}$, for

$h < g$, then H is isogenous to $(\mathrm{Sp}_{2h})^\ell$, for $\ell < g/h$. Since G acts by direct sums of the standard representation, there is some factor which admits a trivial action from H . In other words, the VHS $V_{\mathbb{Q}}$ has a fixed part. As such, it does not vary maximally, so it is forced to once again be isotrivial. $\square$

5. THE MUMFORD–TATE GROUP OF A GENERAL DEFORMATION

Recall that we had $\pi : B' \rightarrow B$ such that $\pi^*V_{\mathbb{C}} = U_1 \oplus \cdots \oplus U_k$, a splitting we now know to occur over $\mathbb{Q}$. Let

$$1 \rightarrow \Gamma' \rightarrow \Gamma \rightarrow \Lambda \rightarrow 1$$

be the corresponding map of monodromy groups, where Γ is the monodromy image of V , Γ' is the monodromy image of π^*V , and Λ is the covering automorphisms of π .

Recall again that $V_{\mathbb{C}}$ is irreducible as a $\mathbb{C}$ -local system, by [Var26]. Since π^*V decomposes as above, the group Λ must transitively permute the irreducible factors U_i . Since Λ is a quotient of Γ , it is generated by local monodromy operators coming from some components $D_i \subseteq D$.

Lemma 5.1. *There exists a component of the discriminant $D_1 \subseteq D$ whose associated local monodromy operator λ acts by a two-cycle on the irreducible factors: for some i, j and all $\ell \neq i, j$,*

$$\lambda : U_i \rightarrow U_j, U_j \rightarrow U_i, U_\ell \xrightarrow{\sim} U_\ell, \quad \lambda^2 = \mathrm{id}.$$

Proof. First notice that there is a natural map $\Lambda \rightarrow \mathfrak{S}_k$, coming from the permutations of the factors $U_1, \dots, U_k$ underlying the action of Λ . In fact, this proof will show that this map is surjective, but it suffices that it surjects onto a subgroup $\bar{\Lambda} \leq \mathfrak{S}_k$. Since Λ is the group of covering automorphisms of π , there is a corresponding intermediate cover

$$B' \xrightarrow{\pi'} B'' \xrightarrow{p} B$$

where $\bar{\Lambda} = \mathrm{Aut}(p)$. Take D_1 to be a component whose local monodromy operator acts by an element of $\bar{\Lambda}$; then λ acts on any U_i by a c_i -cycle; that is,

$$(5.1) \quad \lambda : U_i \rightarrow U_{i_2} \rightarrow \cdots \rightarrow U_{i_c} \rightarrow U_i, \quad \lambda^c : U_i \xrightarrow{\sim} U_i.$$

Now notice that $\bar{\Lambda}$ acts on the limit mixed Hodge structure over a general point of D_1 . In other words, for $b \in D_1$ a general point, we can find a local frame $v_{i_1}, \dots, v_{i_c} \in (F^1\bar{\mathcal{V}}|_{D_1})$ for which

$$\lambda : v_{i_1} \mapsto v_{i_2} \mapsto \cdots \mapsto v_{i_c} \mapsto v_{i_1}.$$

Fix ζ a primitive c -th root of unity, and consider the action of λ on $v_{i_1} + \zeta v_{i_2} + \cdots + \zeta^{c-1} v_{i_c}$; this is an eigenvector in $F^1\bar{\mathcal{V}}|_{D_1}$ with eigenvalue $\bar{\zeta}$. On the other hand, $v_{i_1} + \bar{\zeta} v_{i_2} + \cdots + \bar{\zeta}^{c-1} v_{i_c}$ is another eigenvector living in $F^1\bar{\mathcal{V}}|_{D_1}$, with eigenvalue ζ . By Theorem 3.1, we cannot have two distinct eigenvalues in the F^1 -piece of the limit MHS, so we must have $\zeta = \bar{\zeta} = -1$. In particular, $c = 2$.

Finally, the argument that λ fixes all but two of the U_i is analogous: if not, by the construction above, we could produce two distinct monodromy eigenvectors in the F^1 piece of the limit MHS. This would once again contradict Theorem 3.1. $\square$

We have shown that if the very general fiber of $f : X \rightarrow B$ is not Hodge-generic, then it has general singular fibers whose associated local monodromy group is $\mathbb{Z}/2\mathbb{Z}$. Recall from

the classification of singular fibers of Hwang–Oguiso [HO09] that such a fiber is of Kodaira type I_0^* . Now recall the following result of Lehn:

Proposition 5.2 ([Leh16], Theorem 5.7). *Let X be a hyper-Kähler manifold and $f : X \rightarrow B$ a Lagrangian fibration. Take X' a general deformation of X preserving the fibration, $f' : X' \rightarrow B'$. All general singular fibers of f' are of Kodaira type I_n , II, III, or IV.*

Coupled with our technical result Theorem 5.1, this directly proves the following theorem.

Theorem 5.3. *Let $\mathcal{X}$ be a deformation class of hyper-Kähler manifolds satisfying*

- (1) *there exists $X \in \mathcal{X}$ a hyper-Kähler manifold admitting a Lagrangian fibration,*
- (2) *the second Betti number of any member $X \in \mathcal{X}$ is at least 7.*

Choose $X \in \mathcal{X}$ general among those admitting Lagrangian fibrations, $f : X \rightarrow B$. Then the Mumford–Tate group of the very general fiber X'_b is GSp_{2g} .

Proof. By work of Kim–Laza–Martin [KLM26, Theorem 6.2], f is not isotrivial. Suppose that X_b were not Hodge-generic; then Theorem 5.1 applies, and f has a general singular fiber of type I_0^* , contradicting Theorem 5.2. $\square$

Remark 5.4. Note that the statements of [Leh16, Theorem 5.7] and [KLM26, Theorem 6.2] in their respective papers are somewhat weaker than what is used here, since they only guarantee the existence of an appropriate deformation. However, the proofs of both theorems show that these results hold for the general deformation.

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